



**Joint AEPTX, CPS, Oncor and CenterPoint
Texas 765-kV STEP: Eastern Backbone
Project – ERCOT Independent Review
Status Update**

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RPG Meeting
October 28, 2025

Introduction

- AEP Texas (AEPTX), CPS Energy (CPS), Oncor Electric Delivery and CenterPoint Energy Houston Electric, LLC (CEHE) submitted the Texas 765-kV STEP: Eastern Backbone Project for Regional Planning Group (RPG) review in July 2025
 - This Tier 1 project is estimated to cost \$9.384 billion and will require a Certificate of Convenience and Necessity (CCN) filing
 - Estimated in-service date (ISD) is between 2030 and 2032
 - This project was identified and included in 2024 Regional Transmission Plan (RTP) in Bexar, Dallas, Bell, Kleberg, Rusk, Wharton, other counties in the North Central, South, Central, Southern, Coast, and East Weather Zones
 - This project aligns with the 2024 RTP 765-kV Strategic Transmission Expansion Plan (STEP) Core Plan
- CenterPoint Energy provided an overview presentation and ERCOT provided the study scope at the September RPG Meeting
 - <https://www.ercot.com/calendar/09252025-RPG-Meeting>
- This project is currently under ERCOT Independent Review (EIR)

Recap – Project Proposed by AEPTX, CPS, Oncor and CEHE

- Construct a new Blu Lacy 765/345-kV Substation and Install;
 - Five (5) 765-kV Circuit Breakers in a ring bus configuration expandable to breaker-and-a-half configuration;
 - Four (4) 765-kV, 300 MVA Switch Shunt Reactors (100 MVA/phase);
 - Two (2) Bus Switch Shunts each with a Circuit Breaker; and
 - Two (2) Line Switch Shunts each with a Circuit Breaker
 - Eleven (11) 345-kV Circuit Breakers in a seven-position (7) breaker-and-a-half configuration; and
 - Three (3) 765/345-kV transformers rated to at least 2,403 MVA Normal and 2,772 MVA Emergency;
- Construct a new Watermill 765-kV Switchyard and Install;
 - Twelve (12) Circuit Breakers in a double-bus, double-breaker configuration; and
 - Unspecified reactive device on 765-kV line

Recap – Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- At Existing Watermill 345-kV Substation:
 - Install six (6) Circuit Breakers in a breaker-and-a-half configuration
- At Watermill 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency
- Construct a new Martin Lake 765-kV Switchyard and Install:
 - Eight (8) Circuit Breakers in a double-bus, double-breaker configuration
 - Unspecified reactive device on 765-kV line
- At Existing Martin Lake 345-kV Substation:
 - Install four (4) Circuit Breakers in a double-bus, double-breaker configuration.
- At Martin Lake 765/345-kV Substation Install:
 - Two (2) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency.

Recap – Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Construct a new Hillje 765-kV Substation and Install:
 - Install eight (8) Circuit Breakers in a breaker-and-a-half configuration
- At Existing Hillje 345-kV Substation:
 - Expand breaker-and-a-half configuration by adding two (2) new bays
 - Install five (5) Circuit Breakers in expanded breaker-and-a-half configuration
- At Hillje 765/345-kV Substation Install:
 - Three (3) 765/345-kV transformers, rated to at least 2,403 MVA Normal and 2,772 MVA Emergency

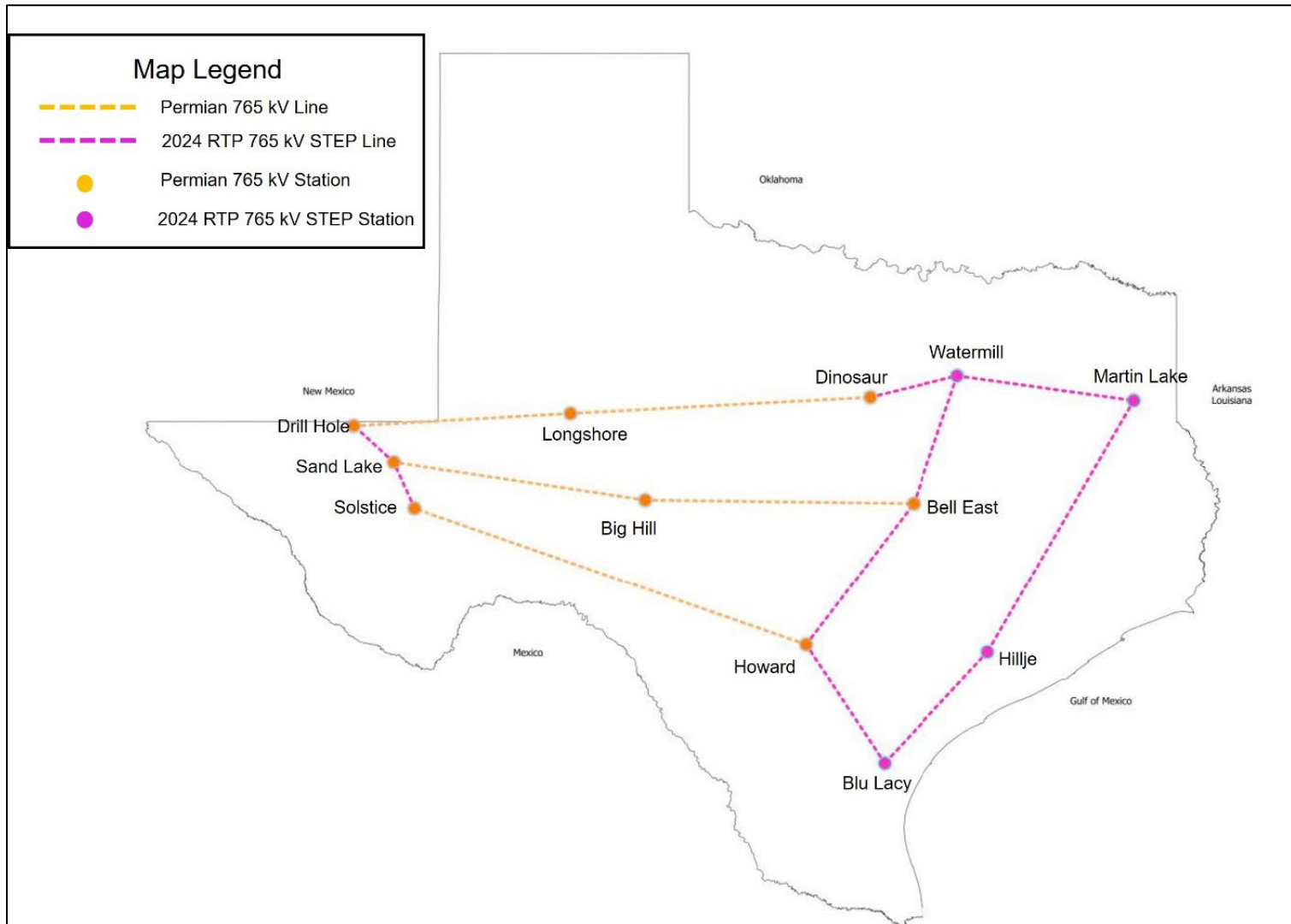
Recap – Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Install two (2) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at New Substation 2 [Dinosaur]
- Install reactive device on 765-kV line at New Substation 2 [Dinosaur]
- Install four (4) 765-kV Circuit Breakers in a double-bus, double-breaker configuration at East Bell Substation
- Install reactive device on 765-kV line at East Bell Substation
- Install four (4) in a Shunt Reactors at Howard Substation double-bus, double-breaker configuration at Howard 765-kV Circuit Breakers Substation
- Install two (2) 300 MVar 765-kV Line
- Install one (1) 300 MVar 765-kV Bus Shunt Reactor at Howard Substation

Recap – Project Proposed by AEPTX, CPS, Oncor and CEHE (Cont.)

- Construct a new Dinosaur to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new right of way (ROW), approximately 78.0-mile;
- Construct a new Watermill to Martin Lake 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 152.4-mile;
- Construct a new Martin Lake to Hillje 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 292.8-mile;
- Construct a new Hillje to Blu Lacy 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 142.8-mile;
- Construct a new Blu Lacy to Howard 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 139.2-mile;
- Construct a new Howard to Bell East 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 170.4-mile; and
- Construct a new Bell East to Watermill 765-kV single-circuit transmission line, with normal and emergency ratings of at least 7,603 MVA, which will require a new ROW, approximately 133.2-mile.

Recap – 765-kV STEP Core Plan Proposed in 2024 RTP



Recap – Study Assumptions and Methodology

- The reliability need of this RPG project was identified and included in the 2024 RTP
- This RPG project aligns with the 2024 RTP 765-kV STEP Core Plan
- ERCOT will utilize the 2024 RTP study for this Independent Review and will assess the need for further analysis
 - Steady-state reliability analysis
 - Congestion Analysis
- ERCOT will perform additional evaluations in accordance with NERC TPL-001-5.1 and ERCOT Planning Criteria
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)

Reliability Analysis

- The 765-kV STEP Core Plan addressed the reliability issues identified in the 2024 RTP
- The 2024 RTP studies highlighted the benefits and cost effectiveness of the 765-kV STEP Core Plan
 - <https://www.ercot.com/calendar/01272025-EHV-765-kV-ERCOT>
 - Steady-state reliability studies
 - Significantly reduce power losses
 - Higher regional transfer capability
 - Sensitivity analysis with a reduced load level (~20 GW less overall load) showed major portions of the 765-kV STEP will still be needed to meet the reduced demand
 - Stability analysis
 - Increase the West Texas Export stability constraint limit
 - Improvement in system strength

Congestion Analysis

- The 2024 RTP studies showed that, in the long-term, 765-kV STEP Core Plan provides greater economic benefit than the traditional 345-kV plan
 - <https://www.ercot.com/calendar/01272025-EHV-765-kV-ERCOT>
 - Estimated in 2025 dollars, the TX 765-kV STEP reduces annual production cost by approximately \$28 million and produces approximately \$229 million more savings in consumer energy cost each year compared to the 345-kV plan in the long term
- Congestion analysis was also performed by using the 2024 RTP 2029 economic case for the 765-kV STEP Core Plan
 - 765-kV STEP Core Plan does not introduce significant new congestion in the study area

Load Scaling Sensitivity Analysis

- Load Scaling Sensitivity Analysis
 - Planning Guide Section 3.1.3(4)(b) requires an evaluation of the potential impact of load scaling on the criteria violations seen in this EIR. Starting 2024, ERCOT RTP adopted a new methodology of having one summer peak case for each study year with non-coincident peaks for each of the Weather Zones, which would eliminate the load scaling impact. The study case did not include load scaling as such load scaling sensitivity analysis is no longer needed

Next Steps and Tentative Timeline

- Next Step
 - Generation Addition Sensitivity
 - Subsynchronous Resonance (SSR) Assessment
- Tentative timeline
 - Status update at November RPG meeting
 - EIR report to be posted in the MIS in November 2025
 - EIR recommendation to TAC in November 2025
 - Seek ERCOT Board of Directors endorsement in December 2025

Thank you!



Stakeholder comments also welcomed through:

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